

Part 1 of 5

Designing with FPGAs:
I²C Master Controller

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Nowadays, field programmable gate arrays (FPGAs) are widely used in a variety of embedded applications as they offer complete system solutions on a single chip including digital signal processing, data storage and processing, software-defined radio, ASIC prototyping, speech recognition, computer vision, cryptography, instrumentation, telecommunications, medical imag-

outside the manufacturing unit. Some of the contemporary FPGA devices offer logic complexity equivalent to that of several million system gates. Also, these devices offer features like built-in hardwired processors and related peripherals, large memory, clock management systems and support for many of the contemporary device-to-device signalling technologies.

In a nutshell, FPGAs are central to any modern electronic system design. FPGAs form a customisable chip flex-

critical information for their day-to-day work. This series of articles is written with an aim to help them interface FPGA with the real-world devices. This is the first part in the series, which gives a brief overview of FPGA essentials in terms of fundamental concepts and other practical information. This is followed by ready-to-use guidelines for designing IC (inter-integrated circuit bus) master controller in an FPGA. The format of each article in the series is such that it provides the introduction to the topic, hardware connection details and VHDL code listing and description. Attempt is made to provide all details related to the topic so as to make the article a complete capsule in itself.

Introduction to FPGA

Internal architecture of an FPGA device has three main parts, namely, array of logic blocks, programmable interconnects and I/O blocks. Fig. 1 shows the architecture of a typical FPGA. Each of the I/O blocks provides individually selectable input, output or bidirectional access to one of the general-purpose I/O pins on the FPGA package. Logic blocks in an FPGA are no more complex than a couple of logic gates or a look-up table feeding a flip-flop. Programmable interconnects not only connect logic blocks to logic blocks but also connect I/O blocks to logic blocks.

Major manufacturers of FPGA devices include Xilinx, Altera, Lattice Semiconductor, MicroSemi (Actel) and SiliconBlue Technologies. Out of these, Xilinx and Altera are the market leaders and manufacture 80 per cent of the FPGAs used in the industry. Xilinx offers Xilinx 7 series (Virtex-7, Kintex-7, Artix-7) based on the 28nm HPL process technology, Zynq-7000 family

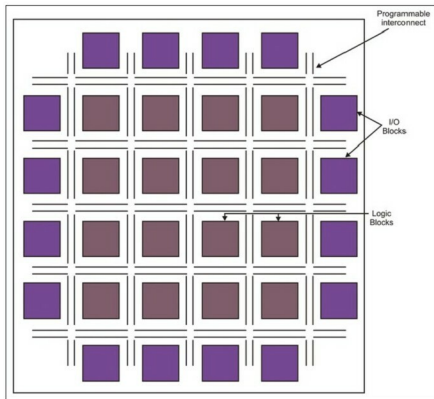


Fig. 1: Architecture of typical FPGA device

ing, defence systems, bioinformatics, computer hardware emulation, re-configurable computing and so on. An FPGA uses an array of logic blocks, which can be configured by the user. The term 'field programmable' signifies that the device is programmable

enough to be deployed in a wide range of applications. They are the backbone of many complex 'system-on-chip' designs.

Designers and application engineers are always in need of a ready reference that provides them all the

